

CSES Graph Algorithms — Problem Reference

All listed tasks: 1 s; 512 MB unless noted. Examples omitted for compactness.

1160 — Planets Queries II

Each planet has exactly one outgoing teleporter t_i . For each query (a, b) , find the minimum number of teleportations needed to reach b from a , or determine that it is impossible. **In:** n q ; destinations $t_1 \dots t_n$; then a b queries

Out: minimum jumps for each query, or -1
Constraints: $1 \leq n, q \leq 2 \cdot 10^5$, $1 \leq t_i, a, b \leq n$.

1192 — Counting Rooms

An $n \times m$ map contains floor $.$ and walls $\#$. Moving in four directions through floor cells, count the connected floor components (rooms).

In: n m ; then n grid rows

Out: number of rooms

Constraints: $1 \leq n, m \leq 1000$.

1193 — Labyrinth

In a grid with walls $\#$, floor $.$, one start A , and one target B , find a shortest four-direction path from A to B . **In:** n m ; then n grid rows
Out: NO, or YES, path length, and any shortest path using L/R/U/D

Constraints: $1 \leq n, m \leq 1000$.

1194 — Monsters

You and all monsters move simultaneously one grid step per turn. Escape from A to any boundary cell without ever occupying a cell at the same time as a monster; the route must be safe even if monsters know it. **In:** n m ; grid with $.$, $\#$, one A , and zero or more M

Out: NO, or YES, a valid path length and D/U/L/R path

Constraints: $1 \leq n, m \leq 1000$; printed path length $\leq nm$.

1195 — Flight Discount

Directed flights have positive prices. Travel from city 1 to city n and use one coupon on a single edge, changing price c to $\lfloor c/2 \rfloor$. Minimize total cost. **In:** n m ; then directed edges a b c

Out: minimum route price

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $c \leq 10^9$; a route exists. Use 64-bit.

1196 — Flight Routes

Find the k cheapest directed routes from city 1 to city n . Routes may revisit vertices, equal-cost routes count separately, and at least k routes exist. **In:** n m k ; then directed edges a b c

Out: k route costs in nondecreasing order

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $c \leq 10^9$, $1 \leq k \leq 10$. Use 64-bit.

1197 — Cycle Finding

Given a directed weighted graph, determine whether it contains a cycle whose total edge weight is negative, and reconstruct one if it exists. **In:** n m ; then directed edges a b c

Out: NO, or YES followed by the cycle vertices in order (start repeated at end)

Constraints: $n \leq 2500$, $m \leq 5000$, $|c| \leq 10^9$. Use 64-bit.

1202 — Investigation

For directed positive-cost flights from 1 to n , determine: minimum route price, number of minimum-price routes modulo $10^9 + 7$, minimum flights among them, and maximum flights among them. **In:** n m ; then directed edges a b c

Out: four integers: distance, count, min edges, max edges

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $c \leq 10^9$; a route exists. Use 64-bit.

1666 — Building Roads

Given an undirected graph, add the minimum number of roads so that every pair of cities becomes connected. Output any set of roads achieving this. **In:** n m ; then undirected edges a b

Out: number k of added roads; then k road endpoints

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$; no loops or duplicate roads.

1667 — Message Route

In an undirected unweighted network, find a route from computer 1 to n using the minimum number of computers (equivalently, shortest number of edges) and reconstruct it. **In:** n m ; then undirected edges a b

Out: IMPOSSIBLE, or route vertex count and any shortest route

Constraints: $2 \leq n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1668 — Building Teams

Split the vertices of an undirected friendship graph into two teams so every edge has endpoints in different teams. **In:** n m ; then undirected edges a b

Out: a label 1 or 2 for each vertex, or IMPOSSIBLE

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1669 — Round Trip

Find any simple cycle in an undirected graph: the route must start/end at the same city and all intermediate cities must be distinct. **In:** n m ; then undirected edges a b

Out: IMPOSSIBLE, or cycle vertex count and vertices in order

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$; no loops or parallel edges.

1671 — Shortest Routes I

In a directed graph with positive edge lengths, find the shortest distance from city 1 to every city. All cities are reachable from 1. **In:** n m ; then directed edges a b c

Out: n distances from city 1

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $c \leq 10^9$. Use 64-bit.

1672 — Shortest Routes II

Given an undirected weighted graph and many queries, answer the shortest distance between each queried pair of cities; pairs may be disconnected. **In:** n m q ; m undirected edges a b c ; then q queries a b

Out: distance per query, or -1 if unreachable

Constraints: $n \leq 500$, $m \leq n^2$, $q \leq 10^5$, $c \leq 10^9$. Use 64-bit.

1673 — High Score

Directed edges may add positive or negative score. Starting at 1, reach n with maximum total score; edges may be reused. If the score can be made arbitrarily large on some valid $1 \rightarrow n$ route, print -1 . **In:** n m ; then directed edges a b x

Out: maximum score, or -1 if unbounded

Constraints: $n \leq 2500$, $m \leq 5000$, $|x| \leq 10^9$; n is reachable from 1. Use 64-bit.

1675 — Road Reparation

Undirected roads have repair costs. Choose roads of minimum total cost so the repaired network is connected; if impossible, report it. **In:** n m ; then undirected roads a b c

Out: minimum total repair cost, or IMPOSSIBLE

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $c \leq 10^9$. Memory 128 MB. Use 64-bit.

1676 — Road Construction

Start with n isolated cities and add m undirected roads one by one. After each addition, report the current number of connected components and the size of the largest component.

In: n m ; then roads a b in insertion order

Out: after each road: component count and largest component size

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1678 — Round Trip II

Find any directed cycle: start/end at the same city and all intermediate cities must be distinct. **In:** n m ; then directed edges a b

Out: IMPOSSIBLE, or cycle vertex count and vertices in order

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1679 — Course Schedule

Given prerequisites $a \rightarrow b$ meaning course a must come before b , output any ordering containing every course that satisfies all requirements. **In:** n m ; then prerequisite edges a b

Out: a valid course order, or IMPOSSIBLE

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1680 — Longest Flight Route

The directed flight graph is a DAG. Find a route from city 1 to city n that visits the maximum possible number of cities, and reconstruct it. **In:** n m ; then directed edges a b

Out: IMPOSSIBLE, or maximum city count and any maximizing route

Constraints: $2 \leq n \leq 10^5$, $m \leq 2 \cdot 10^5$; graph is acyclic.

1681 — Game Routes

A directed acyclic graph has teleporters between levels. Count the number of directed routes from level 1 to level n . **In:** n m ; then directed edges a b

Out: number of routes modulo $10^9 + 7$

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$; graph is acyclic.

1682 — Flight Routes Check

Determine whether a directed graph is strongly connected, i.e. every city can reach every other city. **In:** n m ; then directed edges a b

Out: YES, or NO and any pair a b such that a cannot reach b

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1683 — Planets and Kingdoms

Partition a directed graph into strongly connected components: two planets share a kingdom exactly when each can reach the other.

In: n m ; then directed edges a b

Out: number k of SCCs; then one kingdom label $1..k$ for each planet

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1684 — Giant Pizza

There are m Boolean toppings and n clauses. Each clause contains two literals $+x$ or $-x$; choose topping truth values so at least one literal in every clause is true. **In:** n m ; then n pairs of signed topping literals

Out: one $+/-$ choice for each topping, or IMPOSSIBLE

Constraints: $1 \leq n, m \leq 10^5$, topping indices $1..m$.

1686 — Coin Collector

Each room has k_i coins and directed tunnels connect rooms. You may start and end anywhere and move along tunnels; maximize the total coins obtainable from visited rooms. **In:** n m ; coin values $k_1 \dots k_n$; then directed edges a b

Out: maximum collectible coins

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $k_i \leq 10^9$. Use 64-bit.

1689 — Knight's Tour

Given a starting square on an 8×8 board, construct any knight's tour that visits every square exactly once. **In:** x y , the starting position

Out: an 8×8 grid whose entries $1..64$ give the visit order

Constraints: $1 \leq x, y \leq 8$.

1690 — Hamiltonian Flights

Count directed routes from city 1 to city n that visit every city exactly once. **In:** n m ; then directed edges a b

Out: number of Hamiltonian $1 \rightarrow n$ routes modulo $10^9 + 7$

Constraints: $2 \leq n \leq 20$, $1 \leq m \leq n^2$.

1691 — Mail Delivery

In an undirected graph, find a closed route starting and ending at crossing 1 that uses every street exactly once. **In:** n m ; then undirected edges a b

Out: an Euler circuit as $m + 1$ crossings, or IMPOSSIBLE

Constraints: $2 \leq n \leq 10^5$, $m \leq 2 \cdot 10^5$; no loops or parallel streets.

1692 — De Bruijn Sequence

Construct a minimum-length binary string containing every possible binary string of length n as a contiguous substring. **In:** n

Out: any minimum-length valid bit string

Constraints: $1 \leq n \leq 15$.

1693 — Teleporters Path

In a directed graph, start at level 1, finish at level n , and use every teleporter (edge) exactly once. **In:** n m ; then distinct directed edges a b

Out: an Euler trail as $m + 1$ vertices, or IMPOSSIBLE

Constraints: $2 \leq n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1694 — Download Speed

A directed network edge $a \rightarrow b$ has capacity c . Find the maximum total data flow from server 1 to computer n . **In:** n m ; then directed capacities a b c

Out: maximum flow value

Constraints: $n \leq 500$, $m \leq 1000$, $c \leq 10^9$. Use 64-bit.

1695 — Police Chase

An undirected street graph connects bank 1 to harbor n . Find the minimum number of streets whose removal disconnects them, and output one such minimum cut. **In:** n m ; then undirected streets a b

Out: cut size k ; then the k streets to close

Constraints: $2 \leq n \leq 500$, $m \leq 1000$; at most one street per pair.

1696 — School Dance

Given a bipartite graph of n boys, m girls, and k acceptable pairs, find a maximum matching and output the matched pairs. **In:** n m k ; then acceptable pairs boy girl

Out: matching size r ; then r matched pairs

Constraints: $n, m \leq 500$, $k \leq 1000$.

1711 — Distinct Routes

Directed teleporters may each be used on at most one day. Every day starts at room 1 and ends at room n . Maximize the number of edge-disjoint $1 \rightarrow n$ routes and output them. **In:** n m ; then distinct directed edges a b

Out: maximum route count k ; then for each route its vertex count and vertices

Constraints: $2 \leq n \leq 500$, $m \leq 1000$.

1750 — Planets Queries I

Each planet has one outgoing teleporter t_i . For each query (x, k) , determine the planet reached after exactly k teleportations from x . **In:** n q ; destinations $t_1 \dots t_n$; then queries x k

Out: destination planet for each query

Constraints: $n, q \leq 2 \cdot 10^5$, $0 \leq k \leq 10^9$.

1751 — Planets Cycles

Each planet has one outgoing teleporter. Starting from each planet, keep teleporting until the next planet has already been visited in that walk. Report the number of teleportations performed. **In:** n ; destinations $t_1 \dots t_n$

Out: n answers, one for each starting planet

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq t_i \leq n$.